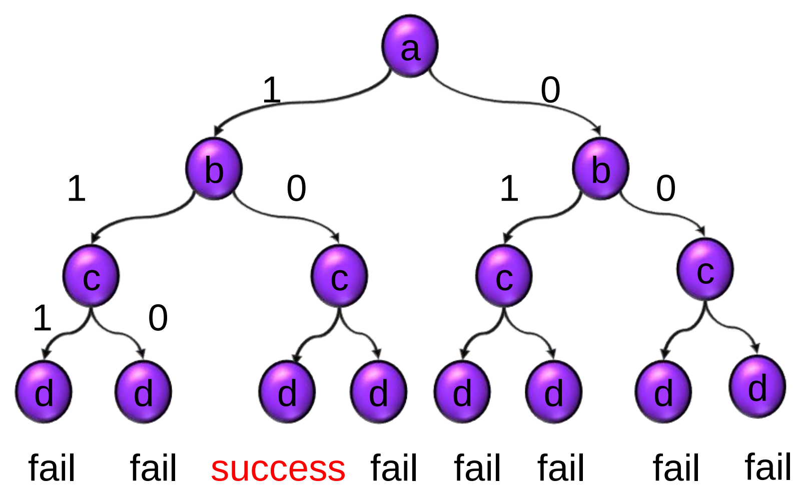
Combinational ATPG

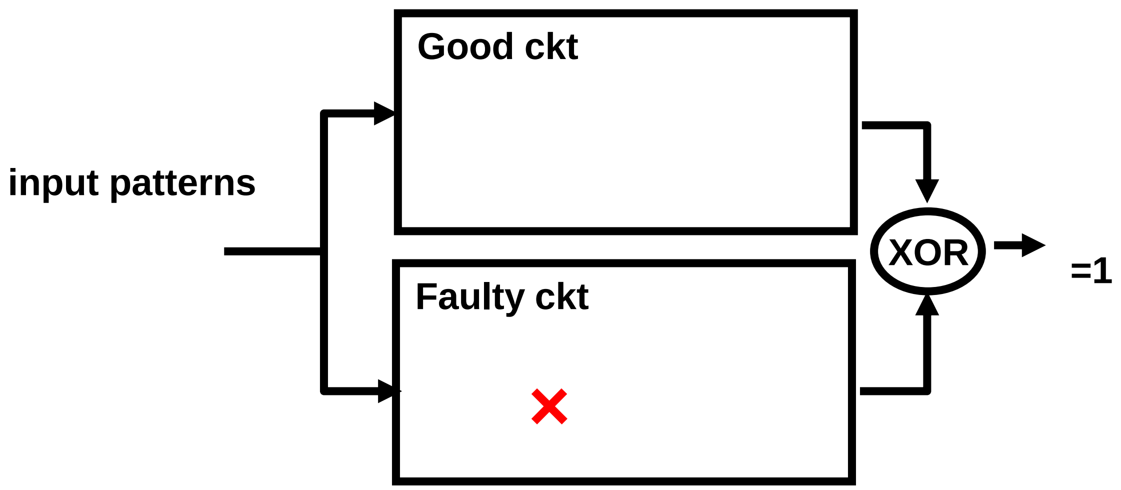
- Introduction
- **Deterministic Test Pattern Generation**
  - Boolean difference *
  - Path sensitization **
  - D-Algorithm [Roth 1966]**
  - PODEM [Goel 1981]**
  - FAN [Fujuwara 1985]**
  - **SAT-based [Larrabee 92] ***
- Acceleration techniques
- Concluding Remarks

*Boolean-based methods
**path-based methods



SAT-based ATPG

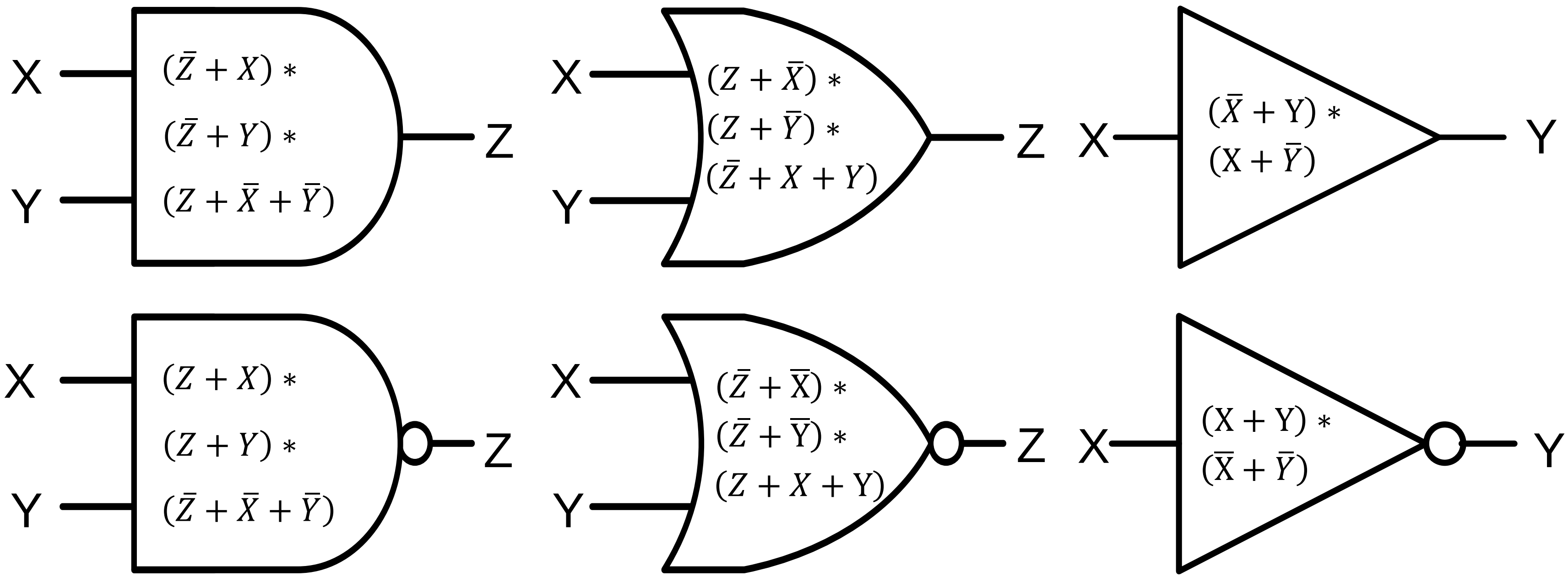
- Idea: convert test generation problem into ***satisfiability problem***
  - Solve for a set of inputs such that
  - (good output) XOR (faulty output) = 1



this is called a ***miter***

Tseitin Transformation [Tseitin 66]

- **Converts circuit into *Conjunctive Normal Form (CNF)***
  - **linear time, linear number of clauses and literals**

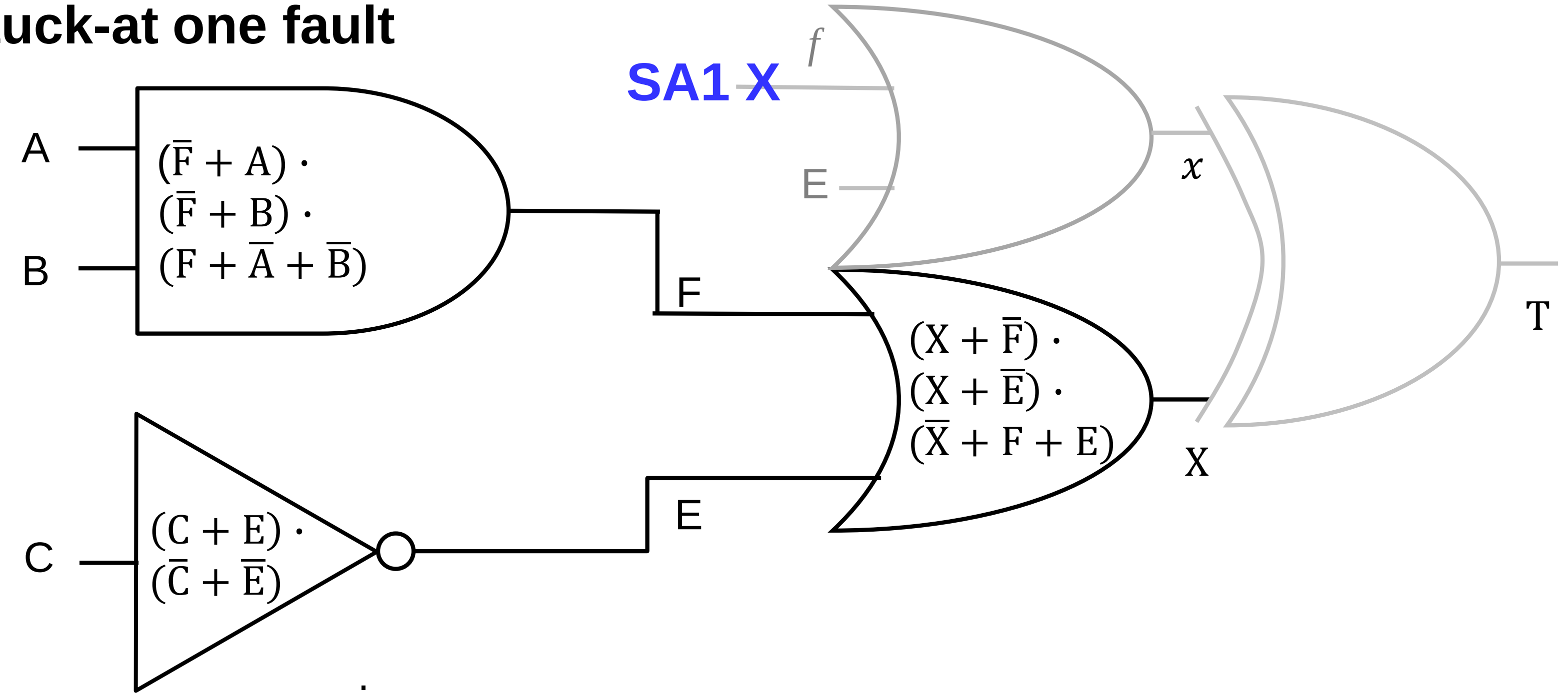


CNF for basic gates

Example (cont'd)

F good
f faulty

- **F stuck-at one fault**



faulty

$$\left(x+\bar{f}\right)\cdot(x+\bar{E})\cdot(\bar{x}+f+E)\cdot(f)$$

good

$$\cdot(C+E)\cdot(\bar{C}+\bar{E})$$
$$\cdot(X+\bar{F})\cdot(X+\bar{E})\cdot(\bar{X}+F+E)$$
$$\cdot(\bar{F}+A)\cdot(\bar{F}+B)\cdot(F+\bar{A}+\bar{B})$$

=1

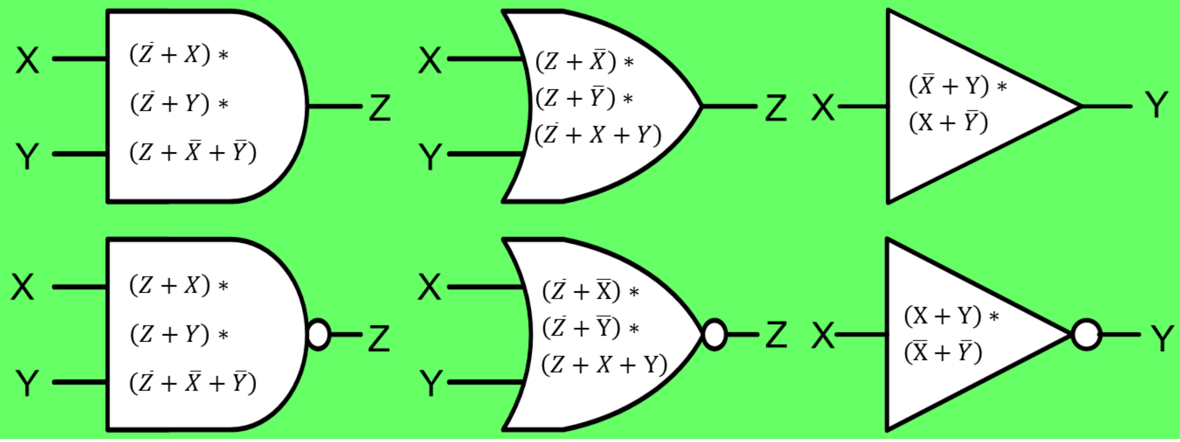
XOR

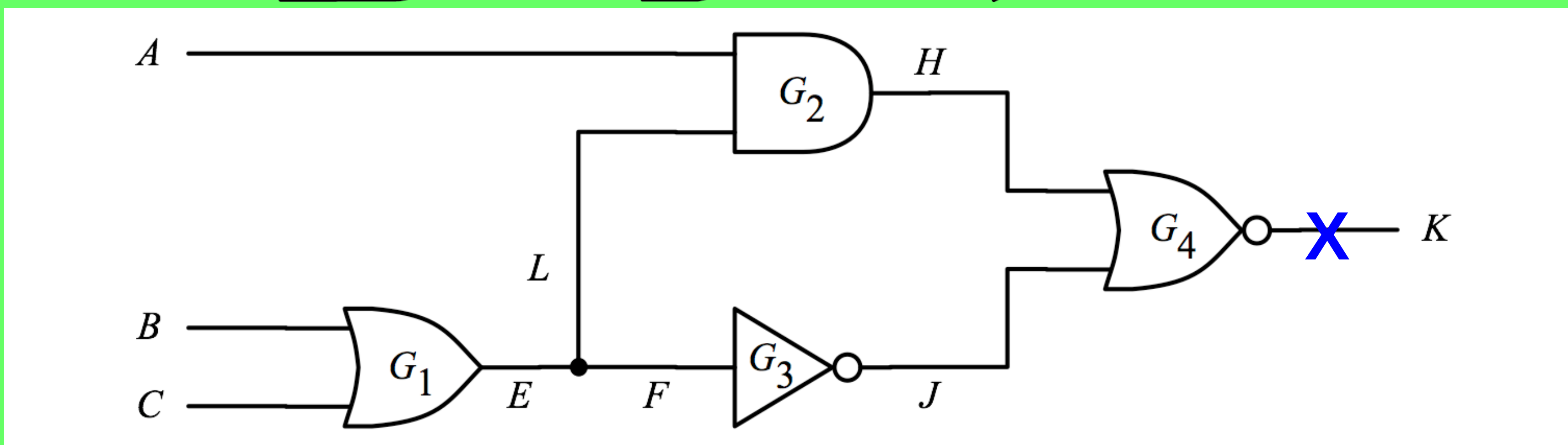
$$\cdot(\bar{X}+x+T)\cdot(X+\bar{x}+T)$$
$$\cdot(\bar{X}+\bar{x}+\bar{T})\cdot(X+x+\bar{T})\cdot T$$

Quiz

Q: Please write CNF to generate a test for K stuck-at one fault

A:





Pros and Cons

- SAT-based ATPG
  - + SAT engine is making big progress recently
  - + proves untestable faults if CNF is unsatisfiable
  - – does not allow don't cares in input
  - – needs to redo CNF every time a target fault is injected
  - – does not preserve circuit structure information
  - – difficult for multi-valued logic (such as high impedance)

SAT-based ATPG Not Used in Commercial Tool